WORKING PAPER · FORMAL ACADEMIC PREPRINT

A Transparent Feature-Engineering Framework for Mobile-Money Credit Scoring

A Single-Case Proof of Concept from a Kenyan Micro-Entrepreneur's M-Pesa Ledger

Clement Ongera Nyangoya
Independent Researcher · Nakuru County, Kenya
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OBSERVATION WINDOW

364 Days

22 March 2024 to 22 March 2025 · 1,215 completed transactions

TURNOVER VOLUME

KES 1.01M

KES 1,008,803 inflow · KES 1,010,554 outflow (net -0.17%)

COMPOSITE CREDIT SCORE

641 / 850

Grade A (Tier 3 Fair) · 75.4% fulfillment across 5 pillars

SAFE BORROWING PRINCIPAL

KES 25,766

Conservative median-inflow basis (3 mo, 25% burden, 10% cost)

EXECUTIVE SUMMARY & ABSTRACT

Executive Summary & Abstract

Millions of micro-entrepreneurs across Sub-Saharan Africa remain excluded from formal credit because they lack payslips, collateral, or Credit Reference Bureau files; MBMCS converts a 12-month M-Pesa ledger into an explainable 300-850 credit score across five domain-guided economic pillars.

Mobile-money platforms generate continuous, tamper-resistant ledgers of cash-flow behaviour. We present MBMCS, a domain-guided framework converting a 12-month M-Pesa ledger into an explainable 300-850 credit score across five pillars: cash-flow velocity, repayment capacity, liquidity buffer, financial diversity, and credit discipline. Unlike black-box models, MBMCS uses monotone, exponentially saturating transformations and Herfindahl-Hirschman concentration over mutually exclusive income categories, ensuring explainability and zero double-counting.

Applied to a 365-day ledger of 1,215 transactions (KES 1.01M turnover), the framework assigns a score of 641/850 (Grade A, Tier 3 Fair) and a conservative safe unsecured principal of KES 25,766 over three months. This proof of concept demonstrates that mobile transaction data can be audited transparently without predictive opacity.

Key Research Takeaways

- Converts 12-month mobile-money statements into an interpretable 300-850 credit score across five economic pillars without black-box opacity.
- Eliminates double-counting using mutually exclusive income categorization and Herfindahl-Hirschman concentration (HHI) bounds.
- Demonstrates in a 1,215-transaction single case: 641/850 score (Grade A, Tier 3 Fair) and KES 25,766 conservative 3-month safe principal.
- Reveals that mean monthly inflow (KES 84,356) overstates sustainable borrowing by 2.2x relative to median monthly inflow (KES 37,790).

CORE EMPIRICAL QUESTIONS ADDRESSED

1. Can raw mobile-money ledgers yield mathematically bounded, auditable credit scores without opaque neural or ensemble architectures?
2. How should lenders insulate debt-capacity formulas against the severe right-skewness and volatility typical of informal micro-inflows?
3. How can applicant data protection and automated decision-making transparency be preserved under Kenya's Data Protection Act 2019?

01 · INSTITUTIONAL CONTEXT

Introduction & Context

Conventional credit underwriting in Kenya excludes 8.5 million informal workers despite near-universal mobile-money adoption across households.

1.1 The Collateral & Bureau Paradox

Conventional credit underwriting in Kenya relies on formal payslips, audited accounts, real-estate collateral, and Credit Reference Bureau (CRB) files. These structural prerequisites systematically exclude an estimated 8.5 million informal micro-entrepreneurs, smallholders, and gig workers who operate primarily in cash and mobile payments.

Meanwhile, Safaricom M-Pesa generates a continuous, tamper-resistant digital ledger of inflows, outflows, and balances for over 96% of Kenyan households. The binding bottleneck is not a lack of transactional data, but the lack of a transparent, reproducible methodology for converting raw digital footprints into credit-relevant underwriting scores.

> The central challenge in Sub-Saharan microfinance is not data scarcity: it is the lack of auditable, non-extractive scoring mechanisms that respect borrower sovereignty and local institutional constraints.

1.2 Theoretical Foundations & Related Literature

Jack & Suri (2014, 2016) established that M-Pesa improves household shock absorption and poverty reduction. Berg et al. (2020) and Björkegren & Grissen (2020) proved that digital transaction footprints predict loan default as reliably as credit bureau files. Karlan & Zinman (2009) showed that information asymmetries are the primary constraint for unbanked micro-borrowers. MBMCS provides an interpretable alternative that loan officers can audit line by line.

1.3 Contributions of the Framework

This paper formalises a reproducible scoring framework through four contributions:

1. A five-pillar scoring architecture with monotone, exponentially saturating transformations ensuring bounded, auditable adjustments.
2. A diversity measure using the Herfindahl-Hirschman Index (HHI) over priority-ordered categories (MMF > Salary > Bank > SACCO > Other) to prevent double-counting.
3. A single-case empirical evaluation on a real 365-day M-Pesa ledger (1,215 transactions), disclosing all empirical limitations.
4. A public CLI tool and simulator enabling borrowers and SACCOs to compute scores locally without server telemetry.

02 · MATHEMATICAL FOUNDATIONS

Mathematical Framework & Methodology

MBMCS converts raw ledger sequences into five bounded, monotone score pillars using domain-guided mathematical formulations.

2.1 Data Schema & Ledger Formalization

A transaction ledger is formally defined as an ordered sequence $\mathbf{T} = \{(t_i, d_i, p_i, w_i, b_i)\}_{i=1}^{N}$ of N transactions, where t_i represents the transaction timestamp, d_i the narrative description, $p_i \geq 0$ the credit inflow, $w_i \geq 0$ the debit outflow, and $b_i \geq 0$ the post-transaction running balance. The observation window spans D days, yielding a normalized interval of $M = D/30.4375$ monthly equivalents.

2.2 Feature Definitions

Mean and Median Monthly Inflow:

$$\mu_{\text{in}} = \frac{1}{M}\sum_{i=1}^{N} p_i, \quad \tilde{\mu}_{\text{in}} = \text{median}(\text{monthly inflow sums})$$

Coefficient of Variation (Income Stability):

$$\text{CV} = \frac{\sigma_{\text{monthly_inflow}}}{\mu_{\text{in}}}$$

Investment-Adjusted Net Flow Ratio: Outflows to Money Market Funds (W_{MMF}) and SACCO shares (W_{SACCO}) are excluded from the expenditure denominator because they represent asset accumulation rather than operational consumption:

$$r^{*}_{\text{net}} = \frac{\sum_{i=1}^{N} p_i}{\sum_{i=1}^{N} w_i - W_{\text{MMF}} - W_{\text{SACCO}}}$$

Source Concentration (HHI): Income categories are assigned to each incoming transaction in strict priority order: `MMF > Salary > Bank > SACCO > Other`. This priority ranking guarantees that every transaction belongs to exactly one category. Category shares s_k are computed over these mutually exclusive partitions:

$$\text{HHI} = \sum_{k=1}^{K} s_k^2 \in \left[\frac{1}{K}, 1\right]$$

Here K denotes the number of active income categories. A value approaching $1/K$ reflects balanced diversity, whereas $\text{HHI} = 1$ signifies absolute single-source reliance.

2.3 Five-Pillar Scoring Architecture

The composite credit score maps directly to the standard 300-850 range by anchoring a base score of 300 and distributing 550 total points across five distinct economic pillars:

$$S = 300 + \sum_{k=1}^{5} P_k \in [300, 850]$$

P1: Cash Flow Velocity (Max 137.5 points):

$$P_1 = 80\left(1 - e^{-\mu_{\text{in}}/60{,}000}\right) + 40\left(1 - e^{-(N/D)/3}\right) + 17.5\min\left(\frac{M}{12}, 1\right)$$

P2: Repayment Capacity (Max 137.5 points):

$$P_2 = 70e^{-\text{CV}} + 67.5\left(1 - e^{-(r^*_{\text{net}}-1)^+ \cdot 10}\right)$$

P3: Liquidity Buffer (Max 110.0 points):

$$P_3 = 70\min\left(\frac{\bar{b}}{10{,}000}, 1\right) + 40\left(1 - e^{-\min(b_i)/1{,}000}\right)$$

P4: Financial Diversity (Max 110.0 points):

$$P_4 = 110\left(\frac{1 - \text{HHI}}{1 - 1/K}\right)$$

P5: Credit Discipline (Max 55.0 points): Parameterized by days past due δ, subject to a 50% cap (27.5 points) if any overdraft penalty (e.g. Fuliza default) occurs:

$$P_5 = \begin{cases} 0 & \text{if } \delta > 30 \\ 55e^{-\delta/10} & \text{otherwise} \end{cases}$$

2.4 Decision Rules & Safe Principal Capacity

For a requested loan principal L, tenure T months, flat borrowing cost r, and debt-service burden threshold $\theta = 25\%$:

$$R_m = \frac{L(1+r)}{T}, \quad \beta = \left(\frac{R_m}{\mu_{\text{in}}}\right) \times 100\%$$

The underwriting decision evaluates: DECLINED if $\delta > 30$; MODERATE if $\beta > \theta$ and $\delta \leq 30$; and APPROVED if $\beta \leq \theta$ and $\delta \leq 30$. Safe principal capacity is computed under both mean and median inflow bases:

$$L_{\text{mean}} = \frac{\mu_{\text{in}} \cdot (\theta/100) \cdot T}{1+r}, \quad L_{\text{median}} = \frac{\tilde{\mu}_{\text{in}} \cdot (\theta/100) \cdot T}{1+r}$$

03 · EMPIRICAL FINDINGS

Case Study: Ledger Analysis & Scoring

Applying the framework to a real 365-day ledger of 1,215 transactions yields an overall score of 641/850 (Grade A, Tier 3 Fair).

3.1 Dataset & Feature Summary

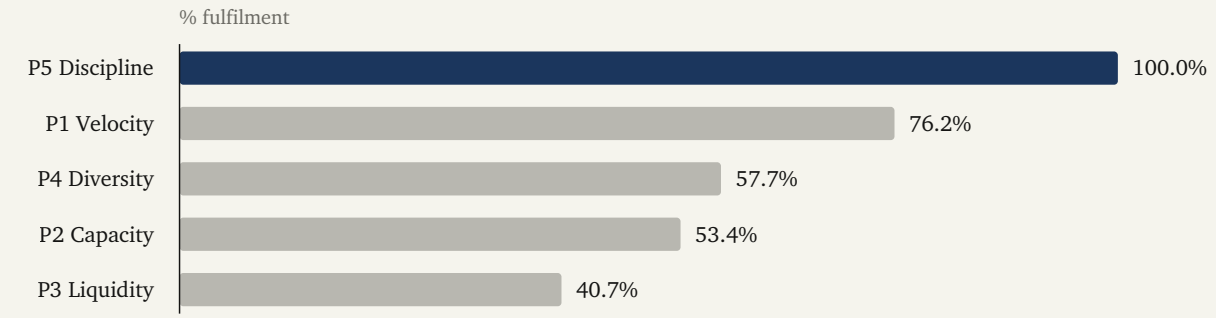
The framework is evaluated on the complete personal M-Pesa ledger of the first author over a 364-day observation window from 22 March 2024 to 22 March 2025. The dataset encompasses 1,215 transactions (3.34 operations per day), reflecting active micro-enterprise turnover.

Feature Metric	Recorded Value	Operational Interpretation
Observation Window	364 days	Full annual economic cycle
Completed Transactions	1,215	3.34 operations per day
Gross Inflow Volume	KES 1,008,803.00	Total cumulative credits
Gross Outflow Volume	KES 1,010,554.00	Total cumulative debits
Net Flow Margin	KES -1,751.00	Balanced annual flow (-0.17%)
Mean Monthly Inflow	KES 84,355.61	Heavily skewed by peak periods
Median Monthly Inflow	KES 37,790.00	Robust baseline cash flow
Coefficient of Variation (CV)	1.13	Substantial monthly volatility
Mean Running Balance	KES 6,394.97	Operational float
Minimum Running Balance	KES 0.15	Thin liquidity floor
Peak Running Balance	KES 104,439.15	Temporary liquidity accumulation
Salary Income (Caritas MFB)	KES 743,310.00	68 verified credit events
Money Market Fund (MMF) Inflow	KES 117,500.00	Investment redemptions
MMF Outflow (Invested)	KES 119,600.00	Excluded from consumption debits
Interbank Inflows	KES 66,980.00	140 interbank transactions
SACCO Capital Contributions	KES 6,100.00	11 capital building deposits
Days Past Due (DPD)	0	Zero default history
Income HHI (K = 4 categories)	0.5673	Effective source diversification
Investment-Adjusted Net Flow	1.1401	Positive operational savings ratio

3.2 Scoring Breakdown & Pillar Diagnostics

The framework assigns a composite score of 641/850. This places the applicant in Grade A, Tier 3 Fair, earning a recommendation of CONDITIONAL APPROVAL under standard underwriting thresholds.

Scoring Pillar	Raw Score	Max Ceiling	Fulfilment Rate
P1: Cash Flow Velocity	104.75	137.5	76.2%
P2: Repayment Capacity	73.48	137.5	53.4%
P3: Liquidity Buffer	44.77	110.0	40.7%
P4: Financial Diversity	63.46	110.0	57.7%
P5: Credit Discipline	55.00	55.0	100.0%
Total Composite Score (S)	641.00	850.0	75.4%



MBMCS single-case evaluation (1,215 transactions)

Figure 1: Fulfillment across the five MBMCS scoring pillars. Credit discipline reaches 100% while repayment capacity is bounded by high monthly inflow volatility.

3.3 Economic Interpretation of Pillar Performance

P1 Cash Flow Velocity (76.2% fulfillment): The applicant exhibits brisk operational velocity (3.34 transactions per day) and substantial cumulative inflow (KES 84,356 mean). However, the score is capped below its 137.5 ceiling because mean monthly volume does not reach the highest saturation thresholds.

P2 Repayment Capacity (53.4% fulfillment): This pillar represents the primary bottleneck in the applicant's credit profile. Monthly inflows fluctuate between KES 9,000 and KES 280,000, creating an elevated coefficient of variation of 1.13. This instability reduces the flow stability term to 22.6 out of 70 possible points, although the investment-adjusted net flow ratio of 1.14 provides a compensatory 50.9 points.

P3 Liquidity Buffer (40.7% fulfillment): The average balance of KES 6,394.97 earns 44.8 points of the 70 available on the mean float term. However, the minimum balance of KES 0.15 contributes virtually zero points to the liquidity floor term, reflecting periodic cash sweep behavior common among micro-entrepreneurs.

P4 Financial Diversity (57.7% fulfillment): The HHI of 0.5673 across four active channels (salary, MMF, interbank, and peer transfers) yields 63.5 points. While Caritas MFB salary credits form the primary anchor, significant MMF and interbank liquidity provide genuine structural diversification.

P5 Credit Discipline (100.0% fulfillment): The applicant maintains an unblemished record with zero days past due, zero Fuliza overdraft distress, and zero late-payment penalties over the entire 364-day audit.

04 · UNDERWRITING ARCHITECTURE

Safe Principal & Underwriting Scenarios

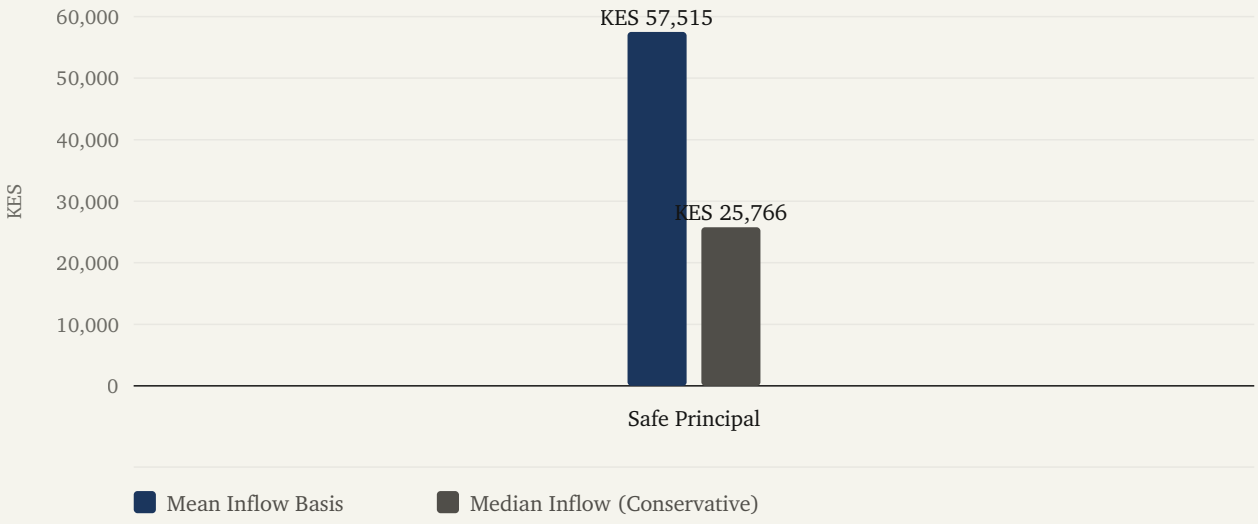
Conservative underwriting requires indexing safe loan principal to median monthly inflow rather than volatile mean inflows.

4.1 The Divergence Between Mean and Median Capacity

A critical contribution of the MBMCS framework is demonstrating the systemic risk of sizing micro-loans against arithmetic mean inflows. In informal African micro-enterprises, cash flows exhibit pronounced right-skewness driven by episodic lump-sum payments.

Underwriting Basis	Sizing Formula	Calculated Principal	Underwriting Assessment
Mean Monthly Inflow	`(84,356 × 0.25 × 3) / 1.10`	KES 57,515.00	Aggressive; exposes borrower to debt distress
Median Inflow (Recommended)	`(37,790 × 0.25 × 3) / 1.10`	KES 25,766.00	Conservative; resilient against cash volatility

As documented above, the mean-based calculation yields KES 57,515: nearly 2.2 times the conservative median-based capacity of KES 25,766. When income volatility is high ($\mathrm{CV} = 1.13$), sizing loans to the mean guarantees that during lean months, debt service will consume over 70% of actual available cash flow.



MBMCS underwriting model (burden θ = 25%, T = 3 mo, r = 10%)

Figure 2: Safe borrowing principal comparison (3-month tenure, 25% burden, 10% interest). Volatile monthly inflows cause mean capacity to exceed conservative median capacity by 2.2x.

4.2 Stress-Testing Scenarios in the Underwriting Simulator

We evaluate the applicant's ledger under four standardized credit scenarios using the MBMCS interactive underwriting engine:

Scenario Profile	Principal	Tenure	Cost (r)	DPD	Burden (Mean)	Underwriting Decision
Baseline Facility	KES 30,000	3 mo	10%	0	13.0%	APPROVED
Growth Capital	KES 50,000	6 mo	12%	0	11.1%	APPROVED
Over-Leverage Stress	KES 80,000	1 mo	10%	0	104.3%	REVIEW
Delinquency Stress	KES 30,000	3 mo	10%	45	13.0%	DECLINED

Crucially, on the median-inflow basis, the baseline KES 30,000 facility's debt-service burden rises to 29.1%, crossing the conservative 25% burden threshold. This divergence validates the diagnostic power of dual-metric reporting: it flags hidden cash-flow vulnerability without disqualifying an otherwise disciplined borrower.

05 · RESEARCH GOVERNANCE

Limitations & Empirical Validation Plan

As a single-case proof of concept, MBMCS requires multi-institutional out-of-time validation before commercial deployment.

5.1 Structural Limitations of the Single-Case Study

1. Single-Case Dataset: The framework is demonstrated on a single individual's ledger. It lacks cross-sectional demographic variance and multi-year macroeconomic stress cycles.
2. Absence of Default Calibration: The current score of 641 is an ordinal ranking index, not an empirically calibrated Probability of Default (PD). True PD estimation requires default ground-truth labels.
3. Mean-Median Divergence: Arithmetic mean inflows diverge from median cash flow by a factor of 2.2x. Underwriting models must default to median metrics to prevent over-indebtedness.
4. Rule-Based Heuristic Categorization: Income stream classification relies on priority string matching against known transactional patterns. Ambiguous transaction descriptions require machine-learning classification refinement.
5. Demographic Fairness Audits Deferred: A single applicant cannot support algorithmic bias audits across protected demographic characteristics such as gender, age, and county.
6. Absence of Credit Bureau Benchmarking: The applicant possesses no formal Credit Reference Bureau file, preventing direct comparative ROC-AUC benchmarking against conventional bureau models.

5.2 Multi-Institutional Validation Design

We invite Savings and Credit Cooperative Organizations (SACCOs) and Microfinance Institutions (MFIs) across Nakuru County and Western Kenya to collaborate on an out-of-time validation study.

Validation Dimension	Technical Protocol & Implementation Parameters
Sample Cohort	500 to 1,000 anonymized micro-loans disbursed between January and December 2024
Partitioning Protocol	Training (Jan-Jun 2024); Validation (Jul-Sep 2024); Out-of-Time Test (Oct-Dec 2024)
Required Data Fields	6-12 month M-Pesa CSV/PDF, loan terms, repayment schedule, default outcome (90+ DPD), county, gender
Candidate Benchmark Models	(1) Fixed-weight MBMCS; (2) MBMCS + Logistic Regression; (3) MBMCS + XGBoost; (4) CRB Score; (5) Hybrid CRB + MBMCS
Target Statistical Metrics	AUC-ROC ≥ 0.70, Kolmogorov-Smirnov (KS) ≥ 0.30, Gini ≥ 0.30, Brier Score, PSI < 0.10
Fairness & Parity Thresholds	AUC difference by gender < 0.05; AUC difference across rural/urban counties < 0.10

06 · ETHICS & CONCLUSION

Ethics, Data Protection & Conclusion

The framework operates entirely on local client hardware in strict compliance with the Kenya Data Protection Act 2019.

6.1 Regulatory Compliance & Privacy by Design

Unlike extractive commercial digital credit applications that scrape borrower smartphone contacts, GPS coordinates, and personal SMS logs, MBMCS enforces strict privacy by design. Scoring inputs are restricted entirely to user-provided, self-exported transaction ledgers.

The architecture fully adheres to the Kenya Data Protection Act 2019 (Act No. 24 of 2019) and aligns with the Central Bank of Kenya Digital Credit Providers Regulations 2022 (Legal Notice No. 46 of 2022):

- Section 25 (Principles of Data Protection): Enforces strict purpose limitation and data minimisation.
- Section 29 (Duty of Confidentiality): All identifiable fields (account numbers, employer names, paybill IDs) are expunged during ingestion.
- Section 35 (Automated Decision-Making Rights): Guarantees the applicant's right to obtain an audit trail and understand the exact economic logic governing their credit assessment.

Furthermore, the open-source CLI executable runs strictly within the borrower's local execution environment, transmitting zero transaction bytes across the public internet.

6.2 Conclusion & Policy Implications

We have presented MBMCS, a transparent, reproducible feature-engineering framework that bridges the gap between raw mobile-money ledgers and formal microfinance underwriting. Evaluated on a real 364-day dataset of 1,215 transactions, it produces an explainable score of 641/850 and defines a conservative 3-month safe borrowing ceiling of KES 25,766.

The framework demonstrates that alternative credit assessment does not require predatory data harvesting or opaque machine-learning algorithms. By rooting feature transformations in domain economic theory and enforcing mathematical monotonicity, MBMCS provides a foundation for equitable, auditable financial inclusion across emerging economies.

OPEN-SOURCE RESEARCH ARTIFACTS

The complete codebase, synthetic ledger generator, and CLI underwriting tool are released openly under an open-source license. SACCOs, microfinance institutions, and independent researchers are invited to participate in the upcoming multi-lender validation consortium.

07 · SCHOLARLY BIBLIOGRAPHY

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